Configuração 8: $M = 128$, $C = 8$, $K = 12$, IoT = 20 dB, QPSK
Perda de SNR do BCD-MMSE (1 iteração) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas
Fundo laranja no rótulo: o modo com o melhor resultado no par, comparando do formato com menos bits para o de mais, sem contar as colunas em que todos os modos ficam abaixo de 0,05 dB
Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo

Perda de SNR (dB)

< 0,1 dB | 0,1 a 0,5 dB | 0,5 a 1 dB | 1 a 2 dB | 2 a 3 dB | 3 a 5 dB | 5 a 10 dB | > 10 dB | × não chega a 10^{-2} | rótulo do melhor modo

TDLA30-5

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.46 1.8%	0.33 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.46 1.8%	0.26 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.46 1.9%	0.33 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.46 1.7%	0.11 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	1.46 1.8%	0.37 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	1.28 2.0%	0.32 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.28 1.8%	0.23 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.28 2.2%	0.32 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.28 1.6%	0.12 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.28 2.0%	0.38 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.40 2.0%	0.33 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.31 1.8%	0.26 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.40 2.1%	0.33 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.27 1.7%	0.17 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.43 2.0%	0.37 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.40 2.0%	0.33 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.31 1.8%	0.26 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.40 2.2%	0.33 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.27 1.6%	0.17 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.43 2.0%	0.37 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.40 1.9%	0.33 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.31 1.8%	0.26 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.40 2.1%	0.33 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.28 1.8%	0.17 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.44 2.0%	0.37 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.40 2.0%	0.33 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.31 1.8%	0.26 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.40 2.2%	0.33 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.27 1.6%	0.17 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.43 2.0%	0.37 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.40 2.0%	0.33 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.31 1.8%	0.26 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.40 2.2%	0.33 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.28 1.8%	0.17 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.44 2.0%	0.37 2.4%	0.00 3.7%	0.00 4.2%

TDLA30-75

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.53 1.8%	0.30 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.53 1.8%	0.24 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	1.53 1.9%	0.30 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	1.53 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	1.53 1.8%	0.32 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	1.18 2.0%	0.31 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.18 1.8%	0.24 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.18 2.2%	0.31 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.18 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.18 2.0%	0.31 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.39 2.0%	0.29 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.35 1.8%	0.21 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.39 2.1%	0.29 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.39 2.0%	0.30 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.39 2.0%	0.29 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.35 1.8%	0.21 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.39 2.2%	0.29 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.39 2.0%	0.30 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.39 1.9%	0.29 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.35 1.8%	0.21 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.39 2.1%	0.29 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.19 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.39 2.0%	0.30 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.39 2.0%	0.29 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.35 1.8%	0.21 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.39 2.2%	0.29 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.39 2.0%	0.30 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.39 2.0%	0.29 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.35 1.8%	0.21 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.39 2.2%	0.29 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.19 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.39 2.0%	0.30 2.4%	0.00 3.7%	0.00 4.2%

TDLA30-300

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.52 1.8%	0.38 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.51 1.8%	0.25 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.52 1.9%	0.38 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.51 1.7%	0.09 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	1.51 1.8%	0.34 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	1.41 2.0%	0.38 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.39 1.8%	0.22 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.41 2.2%	0.38 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.39 1.6%	0.09 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.39 2.0%	0.33 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.45 2.0%	0.39 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.35 1.8%	0.24 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.45 2.1%	0.39 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.24 1.7%	0.09 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.40 2.0%	0.33 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.45 2.0%	0.39 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.35 1.8%	0.24 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.45 2.2%	0.39 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.24 1.6%	0.09 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.40 2.0%	0.33 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.45 1.9%	0.39 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.35 1.8%	0.24 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.45 2.1%	0.39 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.24 1.8%	0.09 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.40 2.0%	0.33 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.45 2.0%	0.39 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.35 1.8%	0.24 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.45 2.2%	0.39 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.24 1.6%	0.09 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.40 2.0%	0.33 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.46 2.0%	0.39 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.36 1.8%	0.24 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.46 2.2%	0.39 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.24 1.8%	0.09 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.41 2.0%	0.33 2.4%	0.00 3.7%	0.00 4.2%

TDLC300-100

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.56 1.8%	0.30 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.56 1.8%	0.19 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	1.56 1.9%	0.30 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	1.56 1.7%	0.09 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	1.56 1.8%	0.31 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	1.58 2.0%	0.30 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.58 1.8%	0.21 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.58 2.2%	0.30 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.58 1.6%	0.09 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.58 2.0%	0.31 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.41 2.0%	0.30 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.38 1.8%	0.23 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.41 2.1%	0.30 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.30 1.7%	0.09 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.42 2.0%	0.31 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.41 2.0%	0.30 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.38 1.8%	0.23 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.41 2.2%	0.30 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.30 1.6%	0.09 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.42 2.0%	0.31 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.41 1.9%	0.30 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.38 1.8%	0.23 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.41 2.1%	0.30 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.30 1.8%	0.09 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.42 2.0%	0.31 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.41 2.0%	0.30 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.38 1.8%	0.23 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.41 2.2%	0.30 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.30 1.6%	0.09 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.42 2.0%	0.31 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.41 2.0%	0.30 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.38 1.8%	0.23 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.41 2.2%	0.30 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.30 1.8%	0.09 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.42 2.0%	0.31 2.4%	0.00 3.7%	0.00 4.2%

TDLB100-400

Quantização	Estouro	<22,10>	<26,10>	<32,12>	<36,12>
AP_TRN	AP_SAT	2.83 1.7%	2.80 2.0%	0.00 3.3%	0.00 3.7%
	AP_SAT_ZERO	3.00 1.6%	2.83 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	2.83 1.8%	2.80 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP	2.92 1.6%	2.89 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	3.11 1.7%	3.08 2.2%	0.00 3.9%	0.00 4.4%
AP_TRN_ZERO	AP_SAT	3.01 1.9%	2.80 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.18 1.7%	2.90 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	3.01 2.1%	2.80 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	3.01 1.4%	2.88 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	3.27 1.9%	3.07 2.3%	0.00 3.9%	0.00 4.4%
AP_RND	AP_SAT	2.80 1.8%	2.80 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.91 1.7%	2.80 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.80 2.0%	2.80 2.4%	0.00 3.9%	0.00 4.4%
	AP_WRAP	2.94 1.6%	2.72 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	3.07 1.9%	3.07 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_ZERO	AP_SAT	2.80 1.9%	2.80 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.91 1.7%	2.80 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.80 2.1%	2.80 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	2.94 1.4%	2.72 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	3.07 1.9%	3.07 2.3%	0.00 3.9%	0.00 4.4%
AP_RND_MIN_INF	AP_SAT	2.80 1.8%	2.80 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.91 1.7%	2.80 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	2.80 2.0%	2.80 2.4%	0.00 3.9%	0.00 4.3%
	AP_WRAP	2.94 1.6%	2.72 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	3.07 1.9%	3.07 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_INF	AP_SAT	2.80 1.9%	2.80 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.91 1.7%	2.80 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.80 2.1%	2.80 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	2.94 1.4%	2.72 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	3.07 1.9%	3.07 2.3%	0.00 3.9%	0.00 4.4%
AP_RND_CONV	AP_SAT	2.80 1.9%	2.80 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.91 1.7%	2.80 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.80 2.1%	2.80 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	2.94 1.6%	2.72 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	3.07 1.9%	3.07 2.3%	0.00 3.9%	0.00 4.4%